Lecture 5
Solutions

Homework 5: Merge them all

Solutions to Homework 5

```
$ bash codeHints/L05/setup_hw5.sh
$ cd practice/hw5
$ git merge drinks
Fast-forward
$ git merge --no-edit desserts
$ git merge --no-edit spicy
$ git merge vegan                    # conflict in menu.txt
$ git status                         # look, then cancel:
$ git merge --abort
$ git merge vegan                    # conflict again, now resolve it
$ cat menu.txt
<<<<<<< HEAD
Pizza (spicy)
=======
Pizza (vegan)
>>>>>>> vegan
Pasta
```

Edit `menu.txt` so that it contains exactly

```
Pizza (spicy, vegan)
Pasta
```

then finish and clean up:

```
$ git add menu.txt
$ git commit --no-edit
$ git branch -d drinks desserts spicy vegan
$ git log --merges --oneline         # three lines
$ printf 'above=spicy\nmerge-commits=3\n' > ../hw5_answers.txt
```

5.1. `main` had no commits of its own beyond the shared start, so Git fast-forwarded: no merge commit, and `main` now points where `drinks` points. If you used `--no-ff`, you get an extra merge commit and the checker reports four merges.

5.2 and 5.3. `main` had moved (it contains `drinks`), so Git made a merge commit each time. `desserts` adds a new file and `spicy` edits `menu.txt`, which `main` has not changed, so there is nothing to resolve.

5.4. `spicy` and `vegan` both changed line 1 of `menu.txt`, which makes a conflict. Between `<<<<<<< HEAD` and `=======` stands *your side*, the branch you are on. Because `spicy` was merged into `main` first, that is the spicy version, which is why the answer to 5.6 is `above=spicy`. After the abort `git status` says `working tree clean`: the merge never happened, and you can try again. Any text that mentions both words on line 1 passes the checker. The line `Pasta` was not part of the conflict and stays unchanged. Check that no marker line is left with `grep -n '<<<<\|====\|>>>>' menu.txt` (it should print nothing).

5.5. `git branch -d` works because every branch is now fully merged. Had Git refused, that would

indicate an unmerged branch.

5.6. Three merge commits: `desserts`, `spicy` and `vegan`. The `drinks` fast-forward made none.

Common mistakes.

- Merging `vegan` before `spicy`. Then `vegan`'s version stands above the divider. The checker notices the wrong order (check "spicy was merged BEFORE vegan") and expects `above=spicy`, so follow the order of the tasks.
- Committing with the marker lines still in the file. Open the file and check.
- Merging while on the wrong branch (for example on `drinks`). Check with `git branch --show-current` before every merge.
- Running `git commit` without `git add menu.txt` first: Git refuses ("Committing is not possible because you have unmerged files").
- Running `git merge --abort` *after* the final commit: it no longer applies. Use `setup_hw5.sh --reset` and start again.